Revision Games Pack — 1.1 Processors, Input, Output & Storage

OCR A Level Computer Science (H446) — Component 01, Section 1.1. Ready-to-print game materials.

Loop cards (dominoes)

How to play: Group size 3–8 (or whole class — deal one or more cards each). Time ~10 min. Cut out the 16 cards and deal them out. One player reads their "Who has…?" clue aloud. Whoever holds the matching term reads "I have: " then reads their own "Who has…?". The chain runs through every card and loops back to the start. If the loop closes, every card has been used correctly.

The chain: **Register** → **Cache** → **Pipelining** → **Core** → **GPU** → **Parallel processing** → **Von Neumann** → **Harvard** → **CISC** → **RISC** → **ALU** → **Control Unit** → **Program Counter (PC)** → **Accumulator (ACC)** → **Virtual memory** → **Volatile** → **(back to Register).**

Card 1 — I have: REGISTER. Who has: the small, fast memory **on or near the CPU** that holds frequently used data to cut slow RAM fetches?

Card 2 — I have: CACHE. Who has: the technique of **overlapping** the fetch, decode and execute of consecutive instructions to raise throughput?

Card 3 — I have: PIPELINING. Who has: an independent processing unit able to run its **own** fetch–decode–execute cycle?

Card 4 — I have: CORE. Who has: the many-core processor that performs the **same operation on large data sets in parallel**?

Card 5 — I have: GPU. Who has: carrying out **multiple computations simultaneously** by splitting a task across cores?

Card 6 — I have: PARALLEL PROCESSING. Who has: the architecture with **one shared memory and bus** for both data and instructions?

Card 7 — I have: VON NEUMANN. Who has: the architecture with **separate memories and buses** for data and instructions, allowing simultaneous access?

Card 8 — I have: HARVARD. Who has: the processor type with **many complex, variable-length** instructions and complexity in hardware (e.g. x86)?

Card 9 — I have: CISC. Who has: the processor type with **few simple fixed-length** instructions aiming for one per cycle, used in ARM mobile devices?

Card 10 — I have: RISC. Who has: the CPU component that performs **arithmetic, logic and shift** operations?

Card 11 — I have: ALU. Who has: the CPU component that **decodes instructions, generates control signals** and synchronises with the clock?

Card 12 — I have: CONTROL UNIT. Who has: the register holding the **address of the next instruction**, incremented each cycle?

Card 13 — I have: PROGRAM COUNTER (PC). Who has: the register that stores the **immediate result of an ALU calculation**?

Card 14 — I have: ACCUMULATOR (ACC). Who has: the use of **secondary storage as extra RAM** when physical RAM is full?

Card 15 — I have: VIRTUAL MEMORY. Who has: the property of memory that **loses its contents when power is removed** (e.g. RAM)?

Card 16 — I have: VOLATILE. Who has: the small, very fast store **inside the CPU** that holds data during processing? *(loops back to Card 1 — REGISTER)*

Taboo cards

How to play: Pairs or two teams. Time ~10 min. One player describes the TERM to their team **without saying any of the four banned words** (or the term itself). One point per correct guess; swap describer each round.

Card 1 — MAR (Memory Address Register) Banned: address · location · memory · read

Card 2 — Cache Banned: fast · CPU · frequently · RAM

Card 3 — Pipelining Banned: overlap · fetch · execute · instructions

Card 4 — Von Neumann Banned: shared · memory · bus · instructions

Card 5 — GPU Banned: parallel · cores · graphics · pixels

Card 6 — Virtual memory Banned: RAM · secondary · disk · pages

Card 7 — Volatile Banned: power · lost · RAM · contents

Card 8 — Optical storage Banned: laser · CD · DVD · pits

Bingo

How to play: Whole class or groups. Time ~10 min. Each player draws a **3×3 grid** and writes **nine** terms chosen from the word bank below. The caller reads clues in random order (tick the ones called). First to a full line (or full grid) shouts "Bingo!" — verify their ticked terms against the caller's list.

Word bank (choose 9):

ALU · Control Unit · Program Counter · MAR · MDR · CIR · Accumulator · Cache · Pipelining · Core · GPU · Clock speed · Von Neumann · Harvard · CISC · RISC · Magnetic storage · Flash/SSD · Optical storage · Virtual memory

Caller's list (read in random order):

1. Performs arithmetic, logic and shift operations. → **ALU**

2. Decodes instructions and generates control signals. → **Control Unit**
3. Holds the address of the next instruction. → **Program Counter**
4. Holds the address of the location to read from or write to. → **MAR**
5. Holds the data or instruction just fetched from memory. → **MDR**
6. Holds the current instruction while it is decoded and executed. → **CIR**
7. Stores the immediate result of an ALU calculation. → **Accumulator**
8. Small fast memory on/near the CPU holding frequently used data. → **Cache**
9. Overlapping fetch, decode and execute to raise throughput. → **Pipelining**
10. An independent unit that runs its own FDE cycle. → **Core**
11. Many-core processor applying one operation across large data sets. → **GPU**
12. Pulses per second that synchronise the CPU, measured in Hz. → **Clock speed**
13. One shared memory and bus for data and instructions. → **Von Neumann**
14. Separate memories and buses for data and instructions. → **Harvard**
15. Many complex, variable-length instructions; complexity in hardware. → **CISC**
16. Few simple fixed-length instructions; used in ARM devices. → **RISC**
17. Stores data as magnetised patterns on a spinning platter. → **Magnetic storage**
18. Non-volatile EEPROM storage with no moving parts. → **Flash/SSD**
19. A laser reads pits and lands via reflected light. → **Optical storage**
20. Uses secondary storage as extra RAM when RAM is full. → **Virtual memory**

Quick-fire "Guess the term"

How to play: Whole class or pairs. Time ~5 min. Read each clue; first correct answer scores. Great as a starter or plenary.

1. The bus that is **unidirectional**, carrying addresses from CPU to memory. → **Address bus**
2. The register pair you must **never swap**: address vs data. → **MAR and MDR**
3. The step in fetch that is **most often forgotten** along with copying MDR→CIR. → **Incrementing the PC**
4. Non-volatile memory holding the **bootstrap/BIOS firmware**. → **ROM**
5. Volatile memory holding **currently running programs and data**. → **RAM**
6. The cache level that is **fastest and smallest**, on the core. → **L1 cache**
7. Excessive paging between RAM and disk that grinds a system to a halt. → **Disk thrashing**
8. The part of a task that **must run sequentially**, limiting the speed-up from adding cores. → **Serial (sequential) fraction**
9. Storage access type of **tape**: you must read through in sequence. → **Serial (sequential) access**
10. Technique flash memory uses to spread writes and prolong cell life. → **Wear levelling**
11. Output device that builds objects **layer by layer**. → **3D printer**
12. The general rule for "choose a device" questions: link each characteristic to the… → **Scenario / requirement (and compare)**